Stratified Hammersley Laws for Luce Permutations

Christopher D. Long*

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Abstract

We prove a Hammersley shape theorem for independent point arrays with exactly one point opportunity on each deterministic vertical fibre. If the density in row i converges uniformly to $q(i/n, \cdot)$ for a continuous profile q, then the complete longest-chain field, divided by $\sqrt{n}$, converges uniformly to the inhomogeneous Hammersley value function

$$\mathcal{U}_q(x,y) = 2 \sup_{\gamma:(0,0)\nearrow(x,y)} \int_0^1 \sqrt{q(\gamma_1(t),\gamma_2(t))\gamma_1'(t)\gamma_2'(t)}\, dt.$$

The convergence holds with fixed-deviation probability $O(e^{-c\sqrt{n}})$, in every finite L^p, and almost surely under arbitrary couplings. We also prove localization of all near-maximal chains around the compact set of maximizing curves.

The stratified theorem yields a general shape law for rankings generated by independent, nonidentically distributed scores. This includes Gaussian Thurstone–Mosteller rankings and, after a bounded compactification of exponential clocks, Luce permutations. For every positive continuous Luce profile we obtain the full permutation shape field, longest increasing and decreasing subsequence laws, robustness under a sublinear set of bounded-rate errors, and stability in the profile. These results establish first-order LIS asymptotics for the stated classes, addressing the question posed by Chatterjee, Diaconis, and Kim and by Borga, Chatterjee, and Diaconis.

Finally, we develop a dyadic corner estimate for endpoint-degenerate rate profiles. It proves that the normalization remains $\sqrt{n}$ under dyadic endpoint comparability, in particular for all regularly varying sampled profiles. As a canonical consequence, Luce permutations with Sukhatme weights $\theta_{n,i} = n-i+1$ satisfy singular increasing- and decreasing-subsequence laws. Reversing the labels gives the corresponding laws for $\theta_{n,i} = i$, with the two variational constants exchanged.

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*galizur@gmail.com

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1 Introduction

The Hammersley process and the longest increasing subsequence of a uniform random permutation are basic models of planar random growth. The limit

$$\frac{\mathrm{LIS}(\pi_n)}{\sqrt{n}} \longrightarrow 2$$

was obtained by Logan–Shepp [13] and, independently, by Vershik–Kerov [20]. See also the Hammersley-process formulation of Aldous–Diaconis [1]. For independent planar samples with a nonconstant density, Deuschel and Zeitouni proved the corresponding variational law and localization of optimal chains [8], and later determined the fixed-deviation large-deviation scales [9]. Related permuton and RSK limits were developed by Sjöstrand [17] and Dubach [10, 11].

For Mallows permutations π_n, with probability proportional to $q_n^{\mathrm{inv}(\pi_n)}$, Mueller and Starr proved a $\sqrt{n}$-scale LIS law when $n(1-q_n) \to \beta \in \mathbb{R}$ [16]. Bhatnagar and Peled proved $\mathrm{LIS}(\pi_n)/(n\sqrt{1-q_n}) \to 1$ when $q_n \to 1$ and $n(1-q_n) \to \infty$ [3].

The Luce model [14] is a natural nonuniform distribution on permutations, obtained by sampling labels without replacement with probabilities proportional to prescribed positive weights. Equivalently, one ranks independent exponential clocks with the corresponding rates. Chatterjee, Diaconis, and Kim asked for the distribution of the longest increasing subsequence [7, Introduction]. Borga, Chatterjee, and Diaconis subsequently established the Luce permuton and local limits and again singled out the LIS problem in their Section 5.3 [5]. The two papers use inverse permutation conventions, but $\mathrm{LIS}(\sigma) = \mathrm{LIS}(\sigma^{-1})$, so the distinction is irrelevant here.

A permuton limit by itself cannot determine the $\sqrt{n}$-scale LIS: a vanishing fraction of planted points can preserve the permuton while creating a much longer monotone subsequence. The clock representation supplies the missing microscopic structure. If $T_{n,i}$ is the clock attached to label i, then exactly

$$\mathrm{LIS}(\sigma_n) = \max\{k : i_1 < \cdots < i_k,\ T_{n,i_1} < \cdots < T_{n,i_k}\}.$$

Thus the terminal LIS is already a longest-chain problem for independent points. The resulting cloud, however, has one point on every deterministic vertical fibre; it is neither an i.i.d. planar sample nor a Poisson cloud. For n independent uniform points in the unit square, the probability of placing exactly one point in each of n equal vertical strips is

$$\frac{n!}{n^n} \sim \sqrt{2\pi n}\, e^{-n}.$$

For a homogeneous Poisson cloud of intensity n on the square, the same event has probability e^{-n}. Conditioning on this event therefore loses a factor $e^{n+o(n)}$, too large to transfer the $e^{-c\sqrt{n}}$ deviation bounds used here. The first purpose of this paper is to prove the appropriate stratified Hammersley theorem directly.

The proof uses a common-core decomposition. Within a thin vertical strip, every fibre law contains a common subdensity. Conditional on selecting this core, the vertical ranks are an exact uniform permutation. The residual subdensities are dominated by a sparse uniform proposal cloud. Concatenating the local chains produces a discrete variational problem, while a finite vertical mesh controls the random block exit heights. This gives the full shape field and quantitative concentration. A restricted version of the same block upper bound proves localization near the maximizing curves.

The second purpose is to isolate the general ranking principle. The reduction does not depend on exponentials: it applies to independent continuous scores whose transformed densities approach a continuous profile after one common increasing compactification. We formulate the induced permuton density and prove the complete shape law. Gaussian random-utility rankings of Thurstone–Mosteller type are an immediate example [19, 15]. The Luce theorem is obtained by specializing this independent-score result to exponential clocks.

The third purpose is to treat vanishing endpoint profiles. A single bounded compactification no longer works when the smallest rate tends to zero. We instead divide the terminal labels into dyadic rate-location blocks. If the rates are comparable inside every such block, the chain supported by the last m labels has exponential moments on the $\sqrt{m}$ scale. Truncating the endpoint and then sending the truncation to zero proves an ordinary $\sqrt{n}$ Hammersley law. Potter bounds show that sampled regularly varying profiles satisfy the dyadic hypothesis [4]. In particular, this gives first-order increasing- and decreasing-subsequence laws for the canonical Sukhatme weights $\theta_{n,i} = n - i + 1$ considered in both Luce papers, and for their label reversal. Fluctuation questions are discussed in Section 12.

1.1 The regular-profile Luce theorem

Let $f \in C[0,1]$ satisfy

$$0 < m_f \le f(x) \le M_f < \infty.$$

For $t \ge 0$, define

$$A_f(t) = \int_0^1 (1 - e^{-f(u)t})\, du, \qquad B_f(t) = A_f'(t) = \int_0^1 f(u) e^{-f(u)t}\, du. \tag{1}$$

Then A_f is a strictly increasing bijection from $[0,\infty)$ to $[0,1)$. Write $\tau_f = A_f^{-1}$ and define

$$\rho_f(x,y) = \frac{f(x)e^{-f(x)\tau_f(y)}}{B_f(\tau_f(y))}, \qquad 0 \le x \le 1, \quad 0 < y < 1. \tag{2}$$

This is the Luce permuton density identified by Borga, Chatterjee, and Diaconis [5, Proposition 2.2].

Let $\mathcal{A}_+(x,y)$ be the set of absolutely continuous coordinatewise nondecreasing paths from $(0,0)$ to (x,y), and put

$$\mathcal{U}_{\rho_f}(x,y) = 2 \sup_{\gamma \in \mathcal{A}_+(x,y)} \int_0^1 \sqrt{\rho_f(\gamma_1(t),\gamma_2(t))\gamma_1'(t)\gamma_2'(t)}\, dt, \tag{3}$$

with endpoint integrals interpreted by monotone truncation when necessary.

A Luce permutation with rates $\theta_{n,1}, \dots, \theta_{n,n}$ is generated by independent clocks

$$T_{n,i} \sim \mathrm{Exp}(\theta_{n,i}), \qquad \sigma_n(i) = \sum_{j=1}^{n} \mathbf{1}_{\{T_{n,j} \le T_{n,i}\}}.$$

For $x, y \in [0,1]$, let $\Lambda_n(x,y)$ be the largest k for which there exist

$$1 \le i_1 < \cdots < i_k \le \lfloor nx \rfloor, \qquad \sigma_n(i_1) < \cdots < \sigma_n(i_k) \le \lfloor ny \rfloor.$$

Thus $\Lambda_n(1,1) = \mathrm{LIS}(\sigma_n)$.

Theorem 1.1 (Regular-profile Luce shape theorem)**.** *Suppose*

$$\theta_{n,i} = a_n\big(f(i/n) + \varepsilon_{n,i}\big), \qquad \max_{1 \le i \le n} |\varepsilon_{n,i}| \longrightarrow 0, \tag{4}$$

where $a_n > 0$ and f is continuous and bounded above and below by positive constants. Then, for every $1 \le p < \infty$,

$$\left\| \sup_{(x,y)\in[0,1]^2} \left| \frac{\Lambda_n(x,y)}{\sqrt{n}} - \mathcal{U}_{\rho_f}(x,y) \right| \right\|_{L^p} \longrightarrow 0. \tag{5}$$

For every $\delta > 0$ there are $c_\delta, C_\delta > 0$ such that

$$\mathbb{P}\left(\sup_{(x,y)\in[0,1]^2} \left| \frac{\Lambda_n(x,y)}{\sqrt{n}} - \mathcal{U}_{\rho_f}(x,y) \right| > \delta \right) \le C_\delta e^{-c_\delta \sqrt{n}} \tag{6}$$

for all sufficiently large n. Consequently the convergence is almost sure under every coupling of the arrays across n.

In particular,

$$\frac{\mathrm{LIS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{L}_+(f) := \mathcal{U}_{\rho_f}(1,1) \tag{7}$$

in every finite L^p and almost surely, and

$$\frac{\mathbb{E}\,\mathrm{LIS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{L}_+(f).$$

The theorem remains valid when the profile approximation fails on a sublinear set of rows.

Theorem 1.2 (Robust profile theorem)**.** *Let $f \in C[0,1]$ be bounded above and below by positive constants. After removing a common rate factor, suppose*

$$0 < \underline{\theta} \le \theta_{n,i} \le \overline{\theta} < \infty \tag{8}$$

uniformly in n, i, and suppose that for every $\eta > 0$,

$$\frac{1}{n}\#\left\{1 \le i \le n : |\theta_{n,i} - f(i/n)| > \eta\right\} \longrightarrow 0. \tag{9}$$

Then all conclusions of Theorem 1.1 remain valid with the same limit $\mathcal{U}_{\rho_f}$.

1.2 Endpoint profiles and the Sukhatme law

For a continuous profile f that is positive on $[0,1)$ and vanishes at 1, the same formula (2) defines a possibly singular permuton density on $(0,1)^2$. Its terminal Hammersley constant is defined by

$$\mathcal{L}_+^{\mathrm{sing}}(f) = \lim_{\delta \downarrow 0} \mathcal{U}_{\rho_f}(1-\delta, 1). \tag{10}$$

The endpoint theorem below gives transparent finite-array hypotheses under which this limit is finite and is the LIS constant.

Corollary 1.3 (Sukhatme weights)**.** *Let σ_n be Luce distributed with*

$$\theta_{n,i} = n - i + 1, \qquad 1 \le i \le n.$$

Then, for every $1 \le p < \infty$,

$$\frac{\mathrm{LIS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{L}_{\mathrm{Suk}} \tag{11}$$

in L^p, with fixed-deviation probability $O(e^{-c\sqrt{n}})$ and almost surely under arbitrary couplings. Here

$$\mathcal{L}_{\mathrm{Suk}} = \mathcal{L}_+^{\mathrm{sing}}(f), \qquad f(x) = 1 - x,$$

and the corresponding density is

$$\rho_{\mathrm{Suk}}(x,y) = \frac{(1-x)\tau(y)^2 e^{x\tau(y)}}{e^{\tau(y)} - \tau(y) - 1}, \tag{12}$$

where τ is the inverse of $A(t) = 1 - (1 - e^{-t})/t$.

Corollary 11.5 also gives the decreasing-subsequence law and the corresponding laws for the reversed rates $\theta_{n,i} = i$. The two constants are exchanged under label reversal. Proposition 11.6 gives a multiplicative truncation bound for both constants, and Proposition 11.7 proves $\mathcal{L}_{\mathrm{Suk}} > 3.0602$ by an explicit trial path.

1.3 Additional consequences

The row-stratified theorem proves more than convergence of lengths. Every near-maximal chain is exponentially likely to lie close to the compact set of maximizing curves; see Theorem 6.3. Vertical reflection gives a longest-decreasing-subsequence law. Let $\mathcal{A}_-(1,1)$ be the class of absolutely continuous paths from $(0,1)$ to $(1,0)$ with first coordinate nondecreasing and second coordinate nonincreasing, and define

$$\mathcal{L}_-(f) = 2 \sup_{\gamma \in \mathcal{A}_-(1,1)} \int_0^1 \sqrt{\rho_f(\gamma_1(t), \gamma_2(t)) \gamma_1'(t)(-\gamma_2'(t))}\, dt, \tag{13}$$

with endpoint actions defined by monotone truncation. If $\kappa_f = M_f/m_f$, then

$$\frac{2}{\sqrt{\kappa_f}} \le \mathcal{L}_+(f), \mathcal{L}_-(f) \le 2\sqrt{\kappa_f}, \qquad \mathcal{L}_+(f)\mathcal{L}_-(f) \ge 1. \tag{14}$$

The constants are locally Lipschitz in the positive profile. These consequences are proved in Section 10.

2 Row-stratified point clouds

2.1 The model and its variational functional

Let $\dagger$ be a cemetery state. For each n, let

$$Y_{n,1}, \dots, Y_{n,n}$$

be independent random variables taking values in $[0,1] \cup \{\dagger\}$. Assume that the restriction of the law of $Y_{n,i}$ to $[0,1]$ has a subdensity $g_{n,i}$; thus

$$g_{n,i} \ge 0, \qquad \int_0^1 g_{n,i}(y)\, dy \le 1,$$

and the remaining mass is placed at $\dagger$. The associated point cloud contains $(i/n, Y_{n,i})$ when $Y_{n,i} \neq \dagger$. We call the deterministic vertical fibre $\{i/n\} \times [0,1]$ row i.

For $x, y \in [0,1]$, define $L_n(x,y)$ to be the largest k for which

$$1 \le i_1 < \cdots < i_k \le \lfloor nx \rfloor, \qquad 0 \le Y_{n,i_1} < \cdots < Y_{n,i_k} \le y.$$

Ties have probability zero under the density assumption.

Let $q \in C([0,1]^2)$ be nonnegative with

$$\int_0^1 q(x,y)\, dy \le 1 \qquad (0 \le x \le 1). \tag{15}$$

For $(x,y) \in [0,1]^2$, define

$$\mathcal{U}_q(x,y) = 2 \sup_{\gamma \in \mathcal{A}_+(x,y)} \int_0^1 \sqrt{q(\gamma_1(t), \gamma_2(t)) \gamma_1'(t) \gamma_2'(t)}\, dt. \tag{16}$$

The value is unchanged by absolutely continuous increasing reparameterizations of the path.

Lemma 2.1 (Basic properties of the value function)**.** *Let $q, \widetilde{q}$ be bounded nonnegative functions on the square. Then*

$$0 \le \mathcal{U}_q(x,y) \le 2\sqrt{\|q\|_\infty\, xy}. \tag{17}$$

Moreover,

$$\sup_{(x,y)\in[0,1]^2} |\mathcal{U}_q(x,y) - \mathcal{U}_{\widetilde{q}}(x,y)| \le 2\left\|\sqrt{q} - \sqrt{\widetilde{q}}\right\|_\infty. \tag{18}$$

If $q \le M$, then for $x \le x'$ and $y \le y'$,

$$0 \le \mathcal{U}_q(x',y') - \mathcal{U}_q(x,y) \le 2\sqrt{M}\left(\sqrt{(x'-x)y'} + \sqrt{x'(y'-y)}\right). \tag{19}$$

In particular, $\mathcal{U}_q$ is uniformly continuous.

Proof. For every admissible path,

$$\int_0^1 \sqrt{\gamma_1'(t)\gamma_2'(t)}\, dt \le \sqrt{\int_0^1 \gamma_1'(t)\, dt}\sqrt{\int_0^1 \gamma_2'(t)\, dt} = \sqrt{xy}.$$

This proves (17). Comparing the two actions along the same path and using the same inequality proves (18).

To prove (19), take a path to (x',y'). Before its first exit from $[0,x]\times[0,y]$, its action is at most $\mathcal{U}_q(x,y)$, since the initial segment can be completed to (x,y) by zero-action horizontal and vertical pieces. If the path exits through the right side, the remaining horizontal increment is at most $x'-x$ and its vertical increment is at most y'. If it exits through the top, the corresponding increments are at most x' and $y'-y$. Applying (17) to the remaining segment and using the sum of the two possible bounds gives (19). □

2.2 The stratified shape theorem

Theorem 2.2 (Row-stratified Hammersley shape theorem)**.** *Assume*

$$\eta_n := \max_{1\le i\le n} \|g_{n,i} - q(i/n,\cdot)\|_\infty \longrightarrow 0, \tag{20}$$

where $q \in C([0,1]^2)$ is nonnegative and satisfies (15). Then, for every $1 \le p < \infty$,

$$\left\| \sup_{(x,y)\in[0,1]^2} \left| \frac{L_n(x,y)}{\sqrt{n}} - \mathcal{U}_q(x,y) \right| \right\|_{L^p} \longrightarrow 0. \tag{21}$$

For every $\delta > 0$ there are $c_\delta, C_\delta > 0$ such that

$$\mathbb{P}\left(\sup_{(x,y)\in[0,1]^2} \left| \frac{L_n(x,y)}{\sqrt{n}} - \mathcal{U}_q(x,y) \right| > \delta \right) \le C_\delta e^{-c_\delta\sqrt{n}} \tag{22}$$

for all sufficiently large n. Consequently the convergence is almost sure under every coupling of the arrays across n.

The proof occupies Sections 3–4. The probabilistic inputs beyond elementary concentration are the classical uniform-permutation LIS law and Talagrand's median inequality.

3 Homogeneous input and a local decomposition

3.1 Uniform permutations

Let H_N denote the LIS length of a uniformly random permutation of $\{1, \dots, N\}$, with $H_0 = 0$. For N independent uniform points in a nondegenerate rectangle, sorting by the first coordinate leaves independent uniform second coordinates. Their relative ranks form a uniform permutation, so the longest-chain length has law H_N. The same rank argument applies to independent identically distributed continuous heights on any N ordered deterministic fibres.

Lemma 3.1 (Uniform-permutation input)**.** *For every $\delta > 0$ there are constants $c_\delta, C_\delta > 0$ such that*

$$\mathbb{P}\left(|H_N - 2\sqrt{N}| > \delta\sqrt{N}\right) \le C_\delta e^{-c_\delta\sqrt{N}} \qquad (N \ge 1). \tag{23}$$

In particular, $H_N/\sqrt{N} \to 2$ in probability.

Proof. Represent H_N as the LIS length of N independent uniform heights. The Logan–Shepp and Vershik–Kerov theorem [13, 20] gives $H_N/\sqrt{N} \to 2$ in probability. Consequently, any median m_N satisfies $m_N/\sqrt{N} \to 2$. Talagrand's median inequalities [18, Theorem 7.1.2] give, for every $u > 0$,

$$\mathbb{P}(H_N \ge m_N + u) \le 2\exp\left\{-\frac{u^2}{4(m_N + u)}\right\},$$
$$\mathbb{P}(H_N \le m_N - u) \le 2\exp\left\{-\frac{u^2}{4m_N}\right\}.$$

For sufficiently large N, $|m_N - 2\sqrt{N}| \le (\delta/2)\sqrt{N}$ and $m_N \le 3\sqrt{N}$. Taking $u = (\delta/2)\sqrt{N}$ proves (23); enlarging C_δ covers the finitely many remaining values of N. □

Sharper nontrivial lower- and upper-tail large-deviation asymptotics have speeds N and $\sqrt{N}$, respectively [9]. The common bound (23) suffices for the arguments below.

Lemma 3.2 (A first-moment chain bound)**.** *Suppose a point cloud is generated by independent row variables, with at most one point in each of r ordered rows, and each row subdensity on an interval J is bounded by D. If R is the longest chain using these rows and heights in J, then*

$$\mathbb{P}(R \ge k) \le \binom{r}{k}\frac{(D|J|)^k}{k!} \le \frac{(rD|J|)^k}{(k!)^2}. \tag{24}$$

Proof. For fixed increasing row indices, the joint density is at most D^k. The region of strictly increasing heights in J^k has volume $|J|^k/k!$. Summing over the $\binom{r}{k}$ possible row sets proves the claim. □

3.2 Common core and residual proposal cloud

Fix an integer $m \ge 1$ and put

$$I_j = \left(\frac{j-1}{m}, \frac{j}{m}\right], \qquad 1 \le j \le m, \qquad \alpha = \frac{1}{m}.$$

Define

$$h_j(y) = \min_{x \in \bar{I}_j} q(x,y), \qquad \omega_m = \sup_{\substack{x,x' \in [0,1],\ |x-x'| \le 1/m \\ y \in [0,1]}} |q(x,y) - q(x',y)|. \tag{25}$$

Then $\omega_m \to 0$ by uniform continuity. Put

$$h_{j,n}(y) = (h_j(y) - \eta_n)_+. \tag{26}$$

If $i/n \in I_j$, then

$$0 \le h_{j,n}(y) \le g_{n,i}(y), \qquad 0 \le g_{n,i}(y) - h_{j,n}(y) \le \omega_m + 2\eta_n. \tag{27}$$

For a vertical interval $J \subset [0,1]$, let $L_{n,j}(J)$ be the longest chain formed by points in $I_j \times J$.

Proposition 3.3 (Local core–residual estimate)**.** *Fix m so large that $\omega_m < 1/2$. For fixed j, J, put*

$$H_j(J) = \int_J h_j(y)\, dy.$$

Then, for every $\delta > 0$, with probability at least $1 - Ce^{-c\sqrt{n}}$ for all sufficiently large n,

$$L_{n,j}(J) \ge 2\sqrt{\alpha n H_j(J)} - \delta\sqrt{n}, \tag{28}$$

$$L_{n,j}(J) \le 2\sqrt{\alpha n H_j(J)} + 2\sqrt{\alpha n \omega_m |J|} + \delta\sqrt{n}. \tag{29}$$

The constants may depend on m, j, J, δ, q but not on n.

Proof. For every row $i/n \in I_j$, mark a point as belonging to the common core in J by prescribing the joint subdensity

$$\mathbb{P}(C_i = 1,\ Y_{n,i} \in dy) = h_{j,n}(y)\mathbf{1}_J(y)\, dy.$$

This is legitimate by (27), and the marking can be performed independently row by row. The core indicators are i.i.d. Bernoulli with parameter

$$p_{j,n}(J) = \int_J h_{j,n}(y)\, dy.$$

Conditional on being in the core, all heights have the same continuous density $h_{j,n}\mathbf{1}_J / p_{j,n}(J)$. Conditional on the number K of core points, their relative vertical order is therefore an exact uniform random permutation of size K. Since the number of rows in I_j is $\alpha n + O(1)$ and $p_{j,n}(J) \to H_j(J)$, Lemma 3.1, combined with a binomial Chernoff bound, gives both upper and lower estimates

$$L^{\mathrm{core}}_{n,j}(J) = 2\sqrt{\alpha n H_j(J)} + o(\sqrt{n})$$

with failure probability $O(e^{-c\sqrt{n}})$. If $H_j(J) = 0$, continuity gives $h_j = 0$ on J, so the core is empty and the same statement holds.

It remains to control residual points in J. By (27), their row subdensities are bounded by

$$d_{m,n} = \omega_m + 2\eta_n.$$

Since $\omega_m < 1/2$ and $\eta_n \to 0$, one has $d_{m,n}|J| \le 1$ for all sufficiently large n. The residual points may therefore be generated by thinning an independent proposal cloud that, in each row, places a

point uniformly in J with probability $d_{m,n}|J|$. Independence is required across rows; the proposal cloud need not be independent of the common core. Conditional on the proposal count, the proposal heights have a uniform random relative order. Lemma 3.1 and a binomial Chernoff bound therefore give

$$L^{\mathrm{res}}_{n,j}(J) \le 2\sqrt{\alpha n \omega_m |J|} + o(\sqrt{n})$$

with failure probability $O(e^{-c\sqrt{n}})$. When $\omega_m = 0$, use Lemma 3.2 and $d_{m,n} \to 0$ instead.

Every chain splits into its core and residual subchains, so

$$L_{n,j}(J) \le L^{\mathrm{core}}_{n,j}(J) + L^{\mathrm{res}}_{n,j}(J),$$

whereas the core itself supplies a lower bound. This proves (28)–(29). □

4 Proof of the stratified shape theorem

4.1 The lower bound

For a nondecreasing partition

$$0 = z_0 \le z_1 \le \cdots \le z_m = 1,$$

define

$$F_m = 2 \sup_{0=z_0\le\cdots\le z_m=1} \sum_{j=1}^{m} \sqrt{\frac{1}{m}\int_{z_{j-1}}^{z_j} h_j(y)\, dy}. \tag{30}$$

Inside column I_j, choose a longest common-core chain in $I_j \times (z_{j-1}, z_j)$. These m chains concatenate because both their horizontal and vertical ranges are ordered. Proposition 3.3, applied to the finitely many rectangles, gives

$$\liminf_{n\to\infty} \frac{L_n(1,1)}{\sqrt{n}} \ge F_m \tag{31}$$

in probability, with fixed-deviation failure probability $O(e^{-c\sqrt{n}})$.

4.2 The upper bound and random exit heights

Take a longest chain and let s_j be the largest height attained by this chain in the first j columns, using $s_j = s_{j-1}$ when column j contributes no point, and setting $s_0 = 0$, $s_m = 1$. Every chain point in column j lies in

$$I_j \times [s_{j-1}, s_j].$$

The intervals are random, so we discretize them. Fix $r \ge 1$ and set

$$J_j^{(r)} = \left[\frac{\lfloor r s_{j-1}\rfloor}{r}, \frac{\lceil r s_j\rceil}{r}\right].$$

For fixed m, r, only finitely many such intervals occur. Proposition 3.3 therefore holds simultaneously for all of them. Writing $M = \|q\|_\infty$ and $\widehat{s}_j = \lceil r s_j\rceil / r$, the intervals $[\widehat{s}_{j-1}, \widehat{s}_j]$ form a genuine partition. The lower overhang of $J_j^{(r)}$ has length at most $1/r$, and hence

$$\sum_{j=1}^{m} 2\sqrt{\frac{1}{m}\int_{J_j^{(r)}} h_j} \le F_m + 2\sqrt{\frac{mM}{r}}. \tag{32}$$

Also

$$\sum_{j=1}^{m} |J_j^{(r)}| \le 1 + \frac{2m}{r}.$$

Thus Cauchy–Schwarz yields

$$\sum_{j=1}^{m} 2\sqrt{\frac{\omega_m}{m}|J_j^{(r)}|} \le 2\sqrt{\omega_m\left(1+\frac{2m}{r}\right)}. \tag{33}$$

Summing the local upper bounds, then letting $r \to \infty$, gives

$$\limsup_{n\to\infty} \frac{L_n(1,1)}{\sqrt{n}} \le F_m + 2\sqrt{\omega_m} \tag{34}$$

in probability, again with fixed-deviation failure probability $O(e^{-c\sqrt{n}})$ for each fixed m.

4.3 Identification of the discrete variational problem

Define a piecewise constant density

$$q_m(x,y) = h_j(y), \qquad x \in I_j. \tag{35}$$

Then

$$\|q - q_m\|_\infty \le \omega_m. \tag{36}$$

Lemma 4.1 (Exact block variational formula)**.** *For every m,*

$$F_m = \mathcal{U}_{q_m}(1,1). \tag{37}$$

Consequently,

$$F_m \longrightarrow \mathcal{U}_q(1,1). \tag{38}$$

Proof. Consider an admissible path and its passage through column I_j. Suppose its vertical coordinate rises there from a to b. Cauchy–Schwarz gives

$$\int_{\gamma\cap(I_j\times[0,1])} \sqrt{h_j(y)x'y'}\; dt \le \sqrt{\frac{1}{m}\int_a^b h_j(y)\; dy}. \tag{39}$$

Summing over columns proves $\mathcal{U}_{q_m}(1,1) \le F_m$.

Conversely, fix a partition $z_0, \dots, z_m$ and put

$$p_j = \int_{z_{j-1}}^{z_j} h_j(y)\; dy.$$

When $p_j > 0$, traverse column j along the absolutely continuous curve specified by

$$x(y) = \frac{j-1}{m} + \frac{1}{m}\frac{\int_{z_{j-1}}^{y} h_j(s)\; ds}{p_j}, \qquad z_{j-1} \le y \le z_j.$$

Equality holds in (39), and the contribution to the action is $2\sqrt{p_j/m}$. If $p_j = 0$, traverse the required horizontal and vertical increments separately; their action is zero. Concatenating the column paths proves the reverse inequality in (37).

Finally, Lemma 2.1 and (36) imply

$$|\mathcal{U}_q(1,1) - \mathcal{U}_{q_m}(1,1)| \le 2\,\|\sqrt{q} - \sqrt{q_m}\|_\infty \le 2\sqrt{\omega_m},$$

which proves (38). □

Combining (31), (34), (38), and $\omega_m \to 0$ proves

$$\frac{L_n(1,1)}{\sqrt{n}} \longrightarrow \mathcal{U}_q(1,1) \tag{40}$$

in probability. Choosing m and then r as functions only of a prescribed fixed deviation δ, and using the exponential estimates in Proposition 3.3, proves the endpoint form of (22).

4.4 The full shape field

Fix $(x,y) \in [0,1]^2$. Apply the preceding argument to the rectangle $[0,x] \times [0,y]$. The last column may have width smaller than $1/m$; treating it as one additional column changes every block value by at most $O(m^{-1/2})$. The lower construction concatenates core chains only inside the rectangle. In the upper construction, the random exit height in each complete column is snapped to the same finite vertical mesh, while the partial final column is bounded by Lemma 3.2. Thus, for every fixed (x,y),

$$\frac{L_n(x,y)}{\sqrt{n}} \longrightarrow \mathcal{U}_q(x,y) \tag{41}$$

with fixed-deviation failure probability $O(e^{-c\sqrt{n}})$.

To pass to uniform convergence, choose a finite square grid of mesh ζ. If (x_-, y_-) and (x_+, y_+) are the lower-left and upper-right grid corners surrounding (x,y), monotonicity gives

$$L_n(x_-, y_-) \le L_n(x,y) \le L_n(x_+, y_+).$$

Lemma 2.1 makes the corresponding oscillation of $\mathcal{U}_q$ uniformly small when ζ is small. A union bound over the fixed grid, followed by the monotonicity sandwich, proves (22) and uniform convergence in probability.

4.5 Moments and almost sure convergence

Let M bound all row densities for sufficiently large n. Applying Lemma 3.2 with $r = n$, $J = [0,1]$, and $D = M$ gives

$$\mathbb{P}(L_n(1,1) \ge k) \le \frac{(nM)^k}{(k!)^2}. \tag{42}$$

Stirling's formula shows that $\{(L_n(1,1)/\sqrt{n})^p : n \ge 1\}$ is uniformly integrable for every finite p. Since the supremum norm error in Theorem 2.2 is bounded by $L_n(1,1)/\sqrt{n} + 2\sqrt{M}$, uniform convergence in probability upgrades to (21). The exponential estimate (22) is summable in n, so Borel–Cantelli proves almost sure convergence under every coupling. This completes the proof of Theorem 2.2.

5 Hamilton–Jacobi interpretation

Proposition 5.1 (Hamilton–Jacobi characterization)**.** *For every nonnegative $q \in C([0,1]^2)$, the value function (16) is the unique continuous, coordinatewise nondecreasing viscosity solution of*

$$(\mathcal{U}_q)_x(\mathcal{U}_q)_y = q \qquad \text{in } (0,1)^2, \tag{43}$$

with incoming boundary values

$$\mathcal{U}_q(x,0) = \mathcal{U}_q(0,y) = 0.$$

No boundary data are prescribed on $\{x=1\} \cup \{y=1\}$.

Proof. Continuity, monotonicity, and the boundary values follow from Lemma 2.1. The viscosity equation follows from [6, Theorem 2.5], applied to the zero extension of $q|_{(0,1)^2}$. This extension does not change the action on the square: segments on its boundary have zero action. Their value function omits the leading factor 2 and solves $U_xU_y = q/4$; multiplying it by 2 gives (43).

For uniqueness, let u, v be two solutions in the stated class. Fix $\varepsilon > 0$ and put

$$w_\varepsilon(x,y) = v(x,y) + \frac{\varepsilon}{1-x} + \frac{\varepsilon}{1-y}, \qquad (x,y) \in [0,1)^2.$$

Every gradient p of a smooth function touching v from below at an interior point has $p_1, p_2 \geq 0$, by coordinatewise monotonicity, and $p_1p_2 \geq q$, by the supersolution property. Consequently every such test for w_ε satisfies

$$\left(p_1 + \frac{\varepsilon}{(1-x)^2}\right)\left(p_2 + \frac{\varepsilon}{(1-y)^2}\right) \geq q(x,y) + \varepsilon^2.$$

Thus w_ε is a strict supersolution. It exceeds u on the incoming axes and tends to infinity at the outgoing edges. If $\sup(u - w_\varepsilon) > 0$, maximize $u(z) - w_\varepsilon(\zeta) - \alpha|z-\zeta|^2/2$. The barrier keeps ζ away from the outgoing edges, and $|z-\zeta| \to 0$ as $\alpha \to \infty$. The strict incoming boundary inequality then keeps both maximizing points in the interior for large α. The two viscosity inequalities, with their common test gradient, imply

$$\varepsilon^2 \leq q(z) - q(\zeta) \longrightarrow 0,$$

a contradiction. Hence $u \leq w_\varepsilon$. Letting $\varepsilon \downarrow 0$, using continuity at the outgoing edges, and interchanging u, v proves uniqueness. This argument requires no strict positivity of q. □

Theorem 2.2 therefore supplies a row-stratified probabilistic approximation to this Hamilton–Jacobi field, with exactly one opportunity for a point in each deterministic row.

6 Localization of near-maximal chains

The finite-block upper bound also locates extremal chains. We work on the compact space Γ of continuous paths $\gamma = (\gamma_1, \gamma_2) : [0,2] \to [0,1]^2$ satisfying

$$\gamma(0) = (0,0), \qquad \gamma(2) = (1,1), \qquad \gamma_1, \gamma_2 \text{ are nondecreasing}, \qquad \gamma_1(s) + \gamma_2(s) = s.$$

Every continuous monotone path from $(0,0)$ to $(1,1)$ has this canonical parameterization after deleting pauses. Each coordinate is 1-Lipschitz. Equip Γ with the metric

$$d_\Gamma(\gamma, \widetilde{\gamma}) = \sup_{0 \leq s \leq 2} \max_{j=1,2} |\gamma_j(s) - \widetilde{\gamma}_j(s)|.$$

Arzelà–Ascoli shows that Γ is compact. Define

$$I_q(\gamma) = 2\int_0^2 \sqrt{q(\gamma(s))\gamma_1'(s)\gamma_2'(s)}\, ds \tag{44}$$

and let

$$\mathscr{M}_q = \{\gamma \in \Gamma : I_q(\gamma) = \mathcal{U}_q(1,1)\}.$$

The supremum over Γ equals $\mathcal{U}_q(1,1)$ by invariance of the action under monotone reparameterization. Distances to subsets of Γ are taken in d_Γ.

Lemma 6.1 (Compactness and an action gap)**.** *If q is continuous and nonnegative, then I_q is upper semicontinuous on Γ. Consequently $\mathscr{M}_q$ is nonempty and compact. For every $\varepsilon > 0$ there is $\eta_\varepsilon > 0$ such that*

$$\sup\{I_q(\gamma) : \operatorname{dist}(\gamma, \mathscr{M}_q) \geq \varepsilon\} \leq \mathcal{U}_q(1,1) - 4\eta_\varepsilon, \tag{45}$$

where the assertion is vacuous if the set on the left is empty.

Proof. For a partition $\Pi : 0 = s_0 < \cdots < s_r = 2$, let $R_\ell(\gamma)$ be the coordinate rectangle spanned by $\gamma(s_{\ell-1})$ and $\gamma(s_\ell)$, put $M_\ell(\gamma) = \max_{R_\ell(\gamma)} q$, and define

$$S_\Pi(\gamma) = 2\sum_{\ell=1}^{r} \sqrt{M_\ell(\gamma)\Delta_\ell\gamma_1\Delta_\ell\gamma_2}.$$

Cauchy–Schwarz gives $I_q(\gamma) \leq S_\Pi(\gamma)$, and refinement can only decrease these upper sums. Along a nested refining sequence with mesh tending to zero, uniform continuity of q and Lebesgue differentiation of γ_1', γ_2' give $S_\Pi(\gamma) \downarrow I_q(\gamma)$. Hence

$$I_q(\gamma) = \inf_\Pi S_\Pi(\gamma).$$

For fixed Π, the map S_Π is continuous, since the maximum of a continuous function over a coordinate rectangle depends continuously on its endpoints. Thus I_q is upper semicontinuous. Compactness gives existence of maximizers and the strict gap away from their maximizing set. □

For a finite increasing chain $\mathcal{C}$, let

$$\Gamma(\mathcal{C}) = \{\gamma \in \Gamma : \mathcal{C} \subseteq \gamma([0,2])\}$$

be its set of monotone interpolants. This set contains the canonical polygonal path $\gamma_\mathcal{C}$ obtained by adjoining $(0,0), (1,1)$ and joining successive vertices by straight segments. For a nonempty closed set $F \subseteq \Gamma$, define

$$M_n(F) = \max\{|\mathcal{C}| : \mathcal{C} \text{ is a chain in the array and } \Gamma(\mathcal{C}) \cap F \neq \varnothing\},$$

with value 0 if there is no such chain. This is a measurable random variable: for each of the finitely many choices of rows, the existence of an interpolant in compact F is a closed condition on the corresponding heights.

Lemma 6.2 (Restricted upper shape bound)**.** *Under the hypotheses of Theorem 2.2, for every nonempty closed $F \subseteq \Gamma$ and every $\delta > 0$ there are $c, C > 0$ such that*

$$\mathbb{P}\left(M_n(F) > \left(\sup_{\gamma \in F} I_q(\gamma) + \delta\right)\sqrt{n}\right) \le Ce^{-c\sqrt{n}} \tag{46}$$

for all sufficiently large n.

Proof. Put $a = \sup_F I_q$ and $M = \|q\|_\infty$. For each $\gamma \in F$, choose a partition Π_γ with

$$S_{\Pi_\gamma}(\gamma) < a + \delta/8.$$

Refine it so that every interval has length less than $(4(M+1))^{-1}$. Continuity gives a neighborhood V_γ on which $S_{\Pi_\gamma} < a + \delta/4$. Compactness of F gives a finite subcover.

Fix one of these neighborhoods and its partition $\Pi : 0 = s_0 < \cdots < s_r = 2$. Suppose $\mathcal{C}$ has an interpolant γ in that neighborhood. Assign each chain vertex (x, y) to the interval containing its canonical parameter $s = x + y$, using $[s_{\ell-1}, s_\ell)$ and making the last interval closed on the right. Every assigned vertex lies in the coordinate rectangle spanned by $\gamma(s_{\ell-1})$ and $\gamma(s_\ell)$.

Snap each lower-left corner down and each upper-right corner up to a fixed square grid of sufficiently small mesh ζ. There are finitely many resulting rectangle sequences. The mesh can be chosen so that every enlarged vertical side has length at most $(2(M+1))^{-1}$ and, uniformly over sequences arising from this neighborhood,

$$2\sum_\ell \sqrt{\left(\sup_{R_\ell} q\right)|\operatorname{proj}_x R_\ell||\operatorname{proj}_y R_\ell|} \le a + \delta/2. \tag{47}$$

This follows from uniform continuity of q and continuity of the upper sums in their finitely many rectangle endpoints.

For one fixed snapped rectangle $R = [u_0, u_1] \times [v_0, v_1]$, put

$$D_n(R) = \sup_R q + \eta_n, \qquad r_n(R) = \#\{i : u_0 \le i/n \le u_1\} = n(u_1 - u_0) + O(1).$$

Both vertical boundaries are included in this row count. For large n, $D_n(R)(v_1 - v_0) \le 1$, so each restricted row law is dominated by a proposal which is present with probability $D_n(R)(v_1 - v_0)$ and is uniform on that interval when present. Conditional on the proposal count, the relative vertical order is a uniform permutation. Lemma 3.1 and a binomial Chernoff bound therefore imply, for every fixed $\xi > 0$,

$$\mathbb{P}\left(L_n(R) > 2\sqrt{nD_n(R)(u_1 - u_0)(v_1 - v_0)} + \xi\sqrt{n}\right) \le C_{R,\xi}e^{-c_{R,\xi}\sqrt{n}}. \tag{48}$$

Here $L_n(R)$ counts chains in the closed rectangle. If $\sup_R q = 0$, the same estimate follows from Lemma 3.2 and $D_n(R) \to 0$. A zero-width rectangle contains at most one row. In all cases the $O(1)$ row-count discrepancy is absorbed by $\xi\sqrt{n}$.

Apply (48) to the finite collection of rectangles, taking the sum of its ξ-errors smaller than $\delta/4$. Each chain is bounded by the sum of the chain lengths in its assigned rectangles. Since $D_n(R) \to \sup_R q$ uniformly over this finite collection, (47) bounds that sum by $(a+\delta)\sqrt{n}$ outside an event of probability $Ce^{-c\sqrt{n}}$. A final union bound over the finite cover proves the result. This argument uses only that the interpolant contains the chain vertices. □

Theorem 6.3 (Localization of near-maximal chains)**.** *Under the hypotheses of Theorem 2.2, for every $\varepsilon > 0$ there are $\eta_\varepsilon, c_\varepsilon, C_\varepsilon > 0$ such that*

$$\mathbb{P}\left(\begin{array}{c} \text{there are a chain } \mathcal{C} \text{ and } \gamma \in \Gamma(\mathcal{C}) \text{ with} \\ |\mathcal{C}| \geq (\mathcal{U}_q(1,1) - \eta_\varepsilon)\sqrt{n}, \qquad \operatorname{dist}(\gamma, \mathscr{M}_q) \geq \varepsilon \end{array} \right) \leq C_\varepsilon e^{-c_\varepsilon \sqrt{n}} \tag{49}$$

for all sufficiently large n. In particular, with the same type of probability bound, every monotone interpolant of every longest chain lies in an ε-neighborhood of $\mathscr{M}_q$. If the maximizing path is unique, these interpolants converge to it in probability and almost surely under arbitrary couplings.

Proof. Let $F_\varepsilon = \{\gamma \in \Gamma : \operatorname{dist}(\gamma, \mathscr{M}_q) \geq \varepsilon\}$. If this set is empty, the assertions are immediate. Otherwise it is compact, and Lemma 6.1 gives $\sup_{F_\varepsilon} I_q \leq \mathcal{U}_q(1,1) - 4\eta_\varepsilon$ for some $\eta_\varepsilon > 0$. Lemma 6.2, with $\delta = \eta_\varepsilon$, bounds every chain admitting an interpolant in F_ε by $(\mathcal{U}_q(1,1) - 3\eta_\varepsilon)\sqrt{n}$, outside the stated exceptional event. This proves (49).

The shape theorem supplies a chain of length at least $(\mathcal{U}_q(1,1) - 2\eta_\varepsilon)\sqrt{n}$ with the same type of failure probability. Thus no longest chain admits an interpolant in F_ε outside the union of these two events. Borel–Cantelli, first for positive rational ε, gives the almost-sure assertion. □

7 Rankings generated by independent scores

Let $X_{n,1}, \ldots, X_{n,n}$ be independent continuous random variables in an interval $I \subseteq \mathbb{R}$, and let $\Psi : I \to (0,1)$ be a fixed increasing homeomorphism. Put $Z_{n,i} = \Psi(X_{n,i})$ and suppose that $Z_{n,i}$ has density $g_{n,i}$. Assume

$$\max_i \|g_{n,i} - q(i/n, \cdot)\|_\infty \longrightarrow 0, \tag{50}$$

where $q \in C([0,1]^2)$ is nonnegative and $\int_0^1 q(x,z)\, dz = 1$ for every x. Define

$$b(z) = \int_0^1 q(x,z)\, dx, \qquad Q(z) = \int_0^z b(s)\, ds. \tag{51}$$

Assume $b(z) > 0$ for $0 < z < 1$. Then Q is an increasing homeomorphism of $[0,1]$, and the induced rank-coordinate density is

$$\rho_q(x,y) = \frac{q(x, Q^{-1}(y))}{b(Q^{-1}(y))}, \qquad 0 < y < 1. \tag{52}$$

Both marginals of ρ_q are uniform. Its actions at the horizontal edges are interpreted by monotone truncation from $0 < y < 1$.

Lemma 7.1 (Vertical coordinate changes)**.** *Let $\mathcal{T}_Q : \Gamma \to \Gamma$ map a path vertically by $(x,z) \mapsto (x, Q(z))$ and then give its image the canonical parameterization. This map is a homeomorphism. If ω_Q is the modulus of continuity of Q, then*

$$d_\Gamma(\mathcal{T}_Q\gamma, \mathcal{T}_Q\widetilde{\gamma}) \leq 2\big(d + \omega_Q(d)\big), \qquad d = d_\Gamma(\gamma, \widetilde{\gamma}). \tag{53}$$

Moreover,

$$I_{\rho_q}(\mathcal{T}_Q\gamma) = I_q(\gamma), \qquad \mathcal{U}_{\rho_q}(x,y) = \mathcal{U}_q(x, Q^{-1}(y)). \tag{54}$$

In particular all these actions are finite, and the rank-coordinate maximizing set is the nonempty compact set

$$\mathscr{M}_{\rho_q} = \{\beta \in \Gamma : I_{\rho_q}(\beta) = \mathcal{U}_{\rho_q}(1,1)\} = \mathcal{T}_Q(\mathscr{M}_q). \tag{55}$$

Proof. For $\gamma \in \Gamma$, the function $h_\gamma(s) = \gamma_1(s) + Q(\gamma_2(s))$ is an increasing homeomorphism of $[0,2]$, and

$$(\mathcal{T}_Q\gamma)(h_\gamma(s)) = (\gamma_1(s), Q(\gamma_2(s))).$$

For two paths at distance d, their transformed points at the same old parameter differ by at most $\max\{d, \omega_Q(d)\}$, and their new parameters differ by at most $d + \omega_Q(d)$. The canonical transformed paths are 1-Lipschitz, proving (53). The inverse is $\mathcal{T}_{Q^{-1}}$, whose continuity follows by the same argument.

The density identity is

$$\rho_q(x, Q(z))Q'(z) = q(x, z). \tag{56}$$

On $\varepsilon \le z \le 1-\varepsilon$, both Q and Q^{-1} are continuously differentiable with bounded derivatives, so the chain rule and reparameterization invariance identify the two path actions on that portion. The two omitted portions of the q-action are bounded by $4\sqrt{\|q\|_\infty\,\varepsilon}$: each has vertical increment at most ε and horizontal increment at most 1. Letting $\varepsilon \downarrow 0$ and using monotone truncation at both horizontal edges proves the first identity in (54). The same argument applies to paths ending at any $(x, Q(z))$, and taking suprema gives its second identity. The homeomorphism and equality of actions then give (55). □

Let

$$\sigma_n(i) = 1 + \#\{j : X_{n,j} < X_{n,i}\}$$

be the label-to-rank permutation, and define its restricted LIS field Λ_n as in the introduction. For an increasing chain $\mathcal{C}$ in its normalized plot, write $\beta_{\mathcal{C}}$ for the canonical polygonal path through its vertices and $(0,0), (1,1)$.

Theorem 7.2 (Independent-score ranking theorem)**.** *Under the hypotheses above, for every $1 \le p < \infty$,*

$$\left\| \sup_{x,y\in[0,1]} \left| \frac{\Lambda_n(x,y)}{\sqrt{n}} - \mathcal{U}_{\rho_q}(x,y) \right| \right\|_{L^p} \longrightarrow 0, \tag{57}$$

with fixed-deviation probability $O(e^{-c\sqrt{n}})$ and almost-sure convergence under arbitrary couplings. In particular,

$$\frac{\mathrm{LIS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{U}_{\rho_q}(1,1)$$

in every such L^p and almost surely, and the normalized expectations have the same limit. For every $\varepsilon > 0$ there are $\eta_\varepsilon, c_\varepsilon, C_\varepsilon > 0$ such that

$$\mathbb{P}\left(\begin{array}{c} \text{there exists a rank chain } \mathcal{C} \text{ with} \\ |\mathcal{C}| \ge (\mathcal{U}_{\rho_q}(1,1) - \eta_\varepsilon)\sqrt{n}, \quad \mathrm{dist}(\beta_{\mathcal{C}}, \mathscr{M}_{\rho_q}) \ge \varepsilon \end{array} \right) \le C_\varepsilon e^{-c_\varepsilon\sqrt{n}} \tag{58}$$

for all sufficiently large n. Every longest-chain polygon localizes around $\mathscr{M}_{\rho_q}$ with the same type of bound. If the maximizing path is unique, these polygons converge to it in probability and almost surely under arbitrary couplings.

Proof. Theorem 2.2 applies to $(i/n, Z_{n,i})$. Let

$$F_n(z) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{Z_{n,i}\le z\}}, \qquad \overline{F}_n(z) = \mathbb{E}F_n(z) = \frac{1}{n}\sum_{i=1}^{n} \int_0^z g_{n,i}(s)\, ds.$$

Uniform density convergence and uniform Riemann-sum approximation give

$$\sup_{z\in[0,1]} |\overline{F}_n(z) - Q(z)| \longrightarrow 0. \tag{59}$$

For fixed $\delta > 0$, choose a deterministic grid whose successive $\overline{F}_n$-increments are at most $\delta/4$. It has $O_\delta(1)$ points. Hoeffding's inequality and monotonicity between grid points yield

$$\mathbb{P}\left(\sup_z |F_n(z) - \overline{F}_n(z)| > \delta\right) \le C_\delta e^{-c_\delta n}. \tag{60}$$

Ties have probability zero, so $\sigma_n(i)/n = F_n(Z_{n,i})$. On $\|F_n - Q\|_\infty \le \delta$, the rank field is therefore sandwiched between score-chain fields cut off at $Q^{-1}((y-\delta-n^{-1})_+)$ and $Q^{-1}((y+\delta)\wedge 1)$. Lemma 7.1, uniform continuity of $\mathcal{U}_q$ and Q^{-1}, and the stratified shape theorem give uniform convergence and the asserted fixed-deviation bound. Bounded row densities give uniform integrability by Lemma 3.2, so the convergence holds in every L^p, $1 \le p < \infty$. Summability gives the almost-sure assertion.

To prove localization, fix $\varepsilon > 0$ and choose $0 < \delta < \varepsilon/4$. For the score vertices of a rank chain $\mathcal{C}$, form the ideal rank polygon $\beta^\circ_{\mathcal{C}}$ through $(i/n, Q(Z_{n,i}))$, with endpoints $(0,0),(1,1)$. On $\|F_n - Q\|_\infty \le \delta$,

$$d_\Gamma(\beta_{\mathcal{C}}, \beta^\circ_{\mathcal{C}}) \le 2\delta \qquad \text{for every chain } \mathcal{C}. \tag{61}$$

Indeed, linear interpolation at corresponding vertices leaves the first coordinate unchanged and changes the second by at most δ; the canonical parameters consequently differ by at most δ. The 1-Lipschitz property of canonical paths gives the stated bound, including after deletion of zero-length segments.

The path $\mathcal{T}_Q^{-1}\beta^\circ_{\mathcal{C}}$ is a monotone interpolant through the exact score vertices. If $\operatorname{dist}(\beta_{\mathcal{C}}, \mathscr{M}_{\rho_q}) \ge \varepsilon$, then (61) places this interpolant in the compact set

$$F = \mathcal{T}_Q^{-1}\{\beta \in \Gamma : \operatorname{dist}(\beta, \mathscr{M}_{\rho_q}) \ge \varepsilon/2\}.$$

This set is disjoint from $\mathscr{M}_q$. If it is empty there is nothing to prove. Otherwise upper semicontinuity and compactness give $\sup_F I_q \le \mathcal{U}_q(1,1) - 4\eta_\varepsilon$ for some $\eta_\varepsilon > 0$. Lemma 6.2, with deviation η_ε, bounds every such chain by $(\mathcal{U}_q(1,1) - 3\eta_\varepsilon)\sqrt{n}$ except with probability $O(e^{-c\sqrt{n}})$. Equations (59) and (60) bound the additional empirical-CDF exception by $O(e^{-c'n})$ for large n. This proves (58). The lower shape bound and Borel–Cantelli give the statements about longest chains and arbitrary couplings. □

7.1 Gaussian Thurstone–Mosteller rankings

The independent-score theorem includes the classical normal random-utility model introduced by Thurstone and developed by Mosteller [19, 15]. Let

$$X_{n,i} = \mu_{n,i} + \sigma G_i, \qquad G_i \overset{\text{i.i.d.}}{\sim} N(0,1), \tag{62}$$

where $\sigma > 0$ and $\max_i |\mu_{n,i} - \mu(i/n)| \to 0$ for a continuous bounded function μ. Choose $\tau > \sigma$ and let H_τ be the distribution function of $N(0,\tau^2)$. Taking $\Psi = H_\tau$, the limiting transformed density is

$$q_\mu(x,z) = \frac{\varphi_\sigma(H_\tau^{-1}(z) - \mu(x))}{\varphi_\tau(H_\tau^{-1}(z))}, \tag{63}$$

where φ_s denotes the $N(0,s^2)$ density.

Corollary 7.3 (Thurstone–Mosteller shape law)**.** *For the ranking generated by* (62)*, the complete LIS field satisfies Theorem 7.2 with* $q = q_\mu$ *and permuton density given by* (52)*.*

Proof. Writing $t = H_\tau^{-1}(z)$, the logarithm of the ratio in (63) has leading term

$$-\frac{1}{2}\left(\frac{1}{\sigma^2} - \frac{1}{\tau^2}\right) t^2 + \frac{\mu(x)}{\sigma^2} t + O(1).$$

Since $\tau > \sigma$ and μ is bounded, $q_\mu(x, z)$ tends uniformly to zero as $z \downarrow 0$ or $z \uparrow 1$. It therefore extends to a bounded continuous density on the closed square. The derivative of the same density ratio with respect to the location parameter is uniformly bounded when that parameter ranges over a fixed compact set. Hence $\max_i |\mu_{n,i} - \mu(i/n)| \to 0$ implies the uniform density convergence required in (50), and Theorem 7.2 applies. □

8 Application to regular Luce profiles

Remove the common factor a_n in (4) and write

$$\lambda_{n,i} = f(i/n) + \varepsilon_{n,i}, \qquad T_{n,i} \sim \mathrm{Exp}(\lambda_{n,i})$$

independently. Since clock ties have probability zero,

$$\mathrm{LIS}(\sigma_n) = \max\left\{k : i_1 < \cdots < i_k, \quad T_{n,i_1} < \cdots < T_{n,i_k}\right\}. \tag{64}$$

Choose

$$0 < c_0 < \frac{m_f}{2} \tag{65}$$

and compactify clock time by

$$Z_{n,i} = 1 - e^{-c_0 T_{n,i}} \in (0, 1). \tag{66}$$

The density of $Z_{n,i}$ is

$$g_{n,i}(z) = \frac{\lambda_{n,i}}{c_0}(1 - z)^{\lambda_{n,i}/c_0 - 1}, \tag{67}$$

and the limiting row density is

$$q_f(x, z) = \frac{f(x)}{c_0}(1 - z)^{f(x)/c_0 - 1}. \tag{68}$$

Because the exponents are uniformly bounded below by a positive constant, q_f is bounded and continuous on the closed square and

$$\max_i \|g_{n,i} - q_f(i/n, \cdot)\|_\infty \longrightarrow 0. \tag{69}$$

The mixture distribution of the compactified score is

$$Q_f(z) = A_f\left(-\frac{1}{c_0}\log(1 - z)\right). \tag{70}$$

Consequently

$$Q_f^{-1}(y) = 1 - e^{-c_0 \tau_f(y)}$$

and a direct substitution into (52) gives

$$\rho_{q_f}(x,y) = \frac{f(x)e^{-f(x)\tau_f(y)}}{B_f(\tau_f(y))} = \rho_f(x,y).$$

Theorem 1.1 is therefore the specialization of Theorem 7.2 to exponential scores.

For later use, note the exact action identity

$$\mathcal{U}_{\rho_f}(x,y) = \mathcal{U}_{q_f}\left(x, 1 - e^{-c_0\tau_f(y)}\right). \tag{71}$$

The endpoint $y = 1$ is obtained by monotone truncation. In particular the possibly unbounded Luce density has a finite Hammersley action whenever the profile is bounded below by a positive constant.

9 Robustness under sparse profile errors

We prove Theorem 1.2. Condition (9), together with uniform boundedness, allows one to choose a deterministic sequence $\delta_n \downarrow 0$ such that

$$E_n = \{i : |\theta_{n,i} - f(i/n)| > \delta_n\} \qquad \text{satisfies} \qquad r_n := |E_n| = o(n). \tag{72}$$

Replace the rates on E_n by $f(i/n)$ and leave all other rates unchanged. The modified array converges uniformly to f and hence satisfies Theorem 1.1.

Couple the original and modified clock arrays so that clocks outside E_n are identical. If two real sequences agree outside a set E, then their LIS lengths L, L' satisfy

$$|L - L'| \le \max\{L(E), L'(E)\}, \tag{73}$$

where $L(E)$ and $L'(E)$ are the LIS lengths restricted to the exceptional indices. Indeed, delete the exceptional entries from a longest subsequence; the remaining subsequence is present in both arrays, and the deleted entries form an increasing subsequence on E.

For this section, decrease the compactification parameter c_0, if necessary, so that

$$0 < c_0 < \frac{1}{2}\min\{m_f, \underline{\theta}\}.$$

All preceding coordinate identities remain valid for this choice. After the compactification (66), both the original and modified exceptional-row densities are then bounded by a constant depending only on $\underline{\theta}, \overline{\theta}, m_f, M_f$, and c_0. Lemma 3.2 therefore yields

$$L(E_n), L'(E_n) = O_{\mathbb{P}}(\sqrt{r_n}) = o_{\mathbb{P}}(\sqrt{n}), \tag{74}$$

and, for every fixed $\delta > 0$, with a bound $O_\delta(e^{-c_\delta\sqrt{n}})$ for the event that either exceptional chain exceeds $\delta\sqrt{n}$.

For the two-parameter permutation fields, changing the r_n exceptional clocks can shift the rank of a nonexceptional label by at most r_n. After deleting exceptional members of a chain, one obtains the deterministic comparison

$$\Lambda_n(x,y) \le \widetilde{\Lambda}_n\left(x, (y + r_n/n) \wedge 1\right) + L(E_n), \tag{75}$$

$$\widetilde{\Lambda}_n(x,y) \le \Lambda_n\left(x, (y + r_n/n) \wedge 1\right) + L'(E_n), \tag{76}$$

where tildes denote the modified array. Uniform continuity of $\mathcal{U}_{\rho_f}$, together with $r_n/n \to 0$ and (74), transfers the full shape theorem, moment convergence, concentration, and almost sure convergence from the modified array to the original one. This proves Theorem 1.2.

Remark 9.1 (Why permuton convergence alone is insufficient). Theorem 1.2 still uses two genuinely microscopic inputs: uniform comparability of the rates and a sublinear exceptional set. In an arbitrary permutation model, one may plant an increasing block of size m_n with $\sqrt{n} \ll m_n \ll n$ without changing the permuton limit. The bounded-density estimate (24) is what rules out this mechanism for bounded-rate Luce rows.

10 Decreasing subsequences, bounds, and stability

10.1 Longest decreasing subsequences

Replace every compactified height $Z_{n,i}$ by $1 - Z_{n,i}$. The transformed row densities still satisfy the hypotheses of Theorem 2.2, and an increasing chain in the reflected coordinates is a decreasing subsequence in the original coordinates. In particular, if $q_f^{\downarrow}(x, z) = q_f(x, 1 - z)$, coordinate invariance gives

$$\mathcal{L}_-(f) = \mathcal{U}_{q_f^{\downarrow}}(1, 1), \tag{77}$$

independently of the admissible compactification parameter. This proves the following statement.

Theorem 10.1 (Longest decreasing subsequences). *Under either the hypotheses of Theorem 1.1 or those of Theorem 1.2,*

$$\frac{\mathrm{LDS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{L}_-(f) \tag{78}$$

with fixed-deviation exponential concentration, almost sure convergence under arbitrary couplings, and convergence in L^p for every $1 \le p < \infty$.

10.2 Comparable-rate bounds

Proposition 10.2 (Universal bounds). *Let $\kappa_f = M_f/m_f$. Then*

$$\frac{2}{\sqrt{\kappa_f}} \le \mathcal{L}_+(f), \mathcal{L}_-(f) \le 2\sqrt{\kappa_f}, \qquad \mathcal{L}_+(f)\mathcal{L}_-(f) \ge 1. \tag{79}$$

Proof. For each fixed $0 < c_0 < m_f$, uniform convergence of the rates eventually keeps $\lambda_{n,i} > c_0$, so the continuous compactification and the coordinate identities above remain valid. The compactified density satisfies

$$q_f(x, z) \le \frac{M_f}{c_0}.$$

Lemma 2.1 gives

$$\mathcal{L}_+(f), \mathcal{L}_-(f) \le 2\sqrt{M_f/c_0}.$$

Letting $c_0 \uparrow m_f$ proves the upper bounds. For the lower bounds, use the row-independent subdensity

$$q_f(x, z) \ge h(z) := \frac{m_f}{c_0}(1 - z)^{M_f/c_0 - 1}, \qquad \int_0^1 h(z)\, dz = \frac{m_f}{M_f}.$$

For any nonnegative continuous h depending only on the vertical coordinate, Cauchy–Schwarz, with equality along $x(z) = \int_0^z h / \int_0^1 h$ when $\int h > 0$, gives

$$\mathcal{U}_h(1, 1) = 2\sqrt{\int_0^1 h(z)\, dz}.$$

Monotonicity of the action proves $\mathcal{L}_+(f) \geq 2/\sqrt{\kappa_f}$, and vertical reflection gives the same bound for $\mathcal{L}_-(f)$. Every deterministic permutation satisfies the Erdős–Szekeres inequality

$$\mathrm{LIS}(\sigma)\,\mathrm{LDS}(\sigma) \geq n$$

[12]. Passing to the limits in Theorems 1.1 and 10.1 yields $\mathcal{L}_+(f)\mathcal{L}_-(f) \geq 1$. □

Corollary 10.3 (Constant profile)**.** *If f is constant, then*

$$\rho_f \equiv 1, \qquad \mathcal{L}_+(f) = \mathcal{L}_-(f) = 2.$$

10.3 Stability in the profile

Proposition 10.4 (Local Lipschitz stability)**.** *Fix $0 < m < M < \infty$. There is $C_{m,M} < \infty$ such that whenever $m \leq f, g \leq M$ are continuous,*

$$|\mathcal{L}_+(f) - \mathcal{L}_+(g)| + |\mathcal{L}_-(f) - \mathcal{L}_-(g)| \leq C_{m,M}\,\|f - g\|_\infty\,. \tag{80}$$

Proof. Choose $c_0 = m/2$. For $a \in [m, M]$ and $z \in [0, 1]$, put

$$s(a, z) = \sqrt{\frac{a}{c_0}}(1 - z)^{(a/c_0 - 1)/2}.$$

Because $(a/c_0 - 1)/2$ is bounded below by a positive constant, both $s(a, z)$ and $s(a, z)|\log(1 - z)|$ are uniformly bounded. Differentiating with respect to a shows

$$\sup_{z \in [0,1]} |s(f(x), z) - s(g(x), z)| \leq C_{m,M}\,|f(x) - g(x)|.$$

Since $s(f(x), z) = \sqrt{q_f(x, z)}$, Lemma 2.1 gives the asserted bound for $\mathcal{L}_+$. Vertical reflection gives the same estimate for $\mathcal{L}_-$. □

11 Endpoint-degenerate Luce profiles

We now allow the limiting rate profile to vanish at the terminal label. The key estimate is independent of any limiting profile and applies to both orientations of a chain.

For $1 \leq m \leq n$, let

$$E_{n,m} = \{n - m + 1, \ldots, n\}$$

be the last m labels. Let $R^+_{n,m}$ and $R^-_{n,m}$ be the lengths of their longest increasing and decreasing clock subsequences, respectively, and write $R_{n,m} = R^+_{n,m}$. For $k \geq 0$, define the dyadic terminal block

$$D_{n,k} = \{i : 2^k \leq n + 1 - i < 2^{k+1}\}. \tag{81}$$

Definition 11.1 (Dyadic endpoint comparability). A triangular array of positive rates has dyadic endpoint constant K if for every n and every nonempty block $D_{n,k}$,

$$\frac{\max_{i \in D_{n,k}} \theta_{n,i}}{\min_{i \in D_{n,k}} \theta_{n,i}} \leq K. \tag{82}$$

Lemma 11.2 (Dyadic corner-chain estimate)**.** *Assume* (82)*. There are constants* $a_K, b_K, C_{p,K} > 0$ *such that, for every* n, m *and either sign,*

$$\mathbb{E}e^{b_K R^{\pm}_{n,m}} \le e^{a_K\sqrt{m}}, \tag{83}$$

$$\mathbb{P}(R^{\pm}_{n,m} \ge u\sqrt{n}) \le \exp\{a_K\sqrt{m} - b_K u\sqrt{n}\}, \qquad u \ge 0, \tag{84}$$

$$\left\|R^{\pm}_{n,m}\right\|_{L^p} \le C_{p,K}\sqrt{m}, \qquad 1 \le p < \infty. \tag{85}$$

One may take $b_K = 1$ *and* $a_K = 2(2+\sqrt{2})\sqrt{eK}$*.*

Proof. Fix a nonempty subset B of one dyadic block, containing r labels. Divide its rates by their minimum. This preserves their relative clock order, and the normalized rates lie in $[1, K]$. The increasing compactification $z = 1 - e^{-t}$ transforms an exponential clock of rate $\lambda \in [1, K]$ into a variable with density

$$\lambda(1-z)^{\lambda-1} \le K, \qquad 0 < z < 1.$$

Write $L^{\pm}_B$ for the two chain lengths within B. Lemma 3.2 gives

$$\mathbb{P}(L^{\pm}_B \ge s) \le \frac{(Kr)^s}{(s!)^2}, \qquad s \ge 1. \tag{86}$$

For the decreasing orientation the proof is identical: the simplex of strictly decreasing heights also has volume $1/s!$. For every $b > 0$, the tail-sum formula therefore gives

$$\begin{aligned}\mathbb{E}e^{bL^{\pm}_B} &= 1 + \sum_{s\ge1}(e^{bs} - e^{b(s-1)})\mathbb{P}(L^{\pm}_B \ge s)\\ &\le \sum_{s\ge0}\frac{(Kre^b)^s}{(s!)^2} \le e^{2\sqrt{Kre^b}}.\end{aligned}$$

The last inequality follows from $\binom{2s}{s} \le 4^s$, by comparison with the even terms of the exponential series.

Put $B_k = D_{n,k} \cap E_{n,m}$ and omit empty intersections. For either fixed sign the variables $L^{\pm}_{B_k}$ are independent, and every chain in $E_{n,m}$ splits into chains in these sets. Hence

$$R^{\pm}_{n,m} \le \sum_k L^{\pm}_{B_k}.$$

If $J = \lfloor \log_2 m \rfloor$, then $|B_k| \le 2^k$ and only $0 \le k \le J$ occur. Consequently

$$\sum_k \sqrt{|B_k|} \le \sum_{k=0}^{J} 2^{k/2} \le (2+\sqrt{2})\sqrt{m}.$$

Multiplying the exponential-moment bounds with $b = 1$ proves (83) with the stated constants. Markov's inequality gives (84); integrating its minimum with 1 proves (85), since $m \ge 1$. □

Let $f \in C[0,1]$ satisfy

$$f(x) > 0 \quad (0 \le x < 1), \qquad f(1) = 0. \tag{87}$$

Define A_f, B_f, τ_f and ρ_f by (1)–(2). These formulas remain valid: $f > 0$ almost everywhere implies $A_f(t) \uparrow 1$, and ρ_f has uniform marginals on the open square. Set

$$\rho_f^+(x,y) = \rho_f(x,y), \qquad \rho_f^-(x,y) = \rho_f(x,1-y).$$

The reflected density converts decreasing chains into increasing chains. For these possibly singular densities, endpoint actions are interpreted by monotone truncation in clock time, as in the following identity.

For $0 < \delta < 1$, put

$$f_\delta(s) = f((1-\delta)s), \qquad 0 \le s \le 1, \tag{88}$$

and define

$$\ell_f^\pm(\delta) = \sqrt{1-\delta}\,\mathcal{L}_\pm(f_\delta), \qquad \ell_f(\delta) = \ell_f^+(\delta). \tag{89}$$

Then

$$\ell_f^\pm(\delta) = \mathcal{U}_{\rho_f^\pm}(1-\delta, 1). \tag{90}$$

To justify this identity, first restrict the raw clock coordinate to $0 \le t \le T < \infty$. On this interval the transformation $y = A_f(t)$ and its inverse are continuously differentiable, with strictly positive derivatives. The density transformation gives equality of the raw-clock action and the rank-coordinate action. Rescaling $x = (1-\delta)u$ gives the factor $\sqrt{1-\delta}$ on the left. For the decreasing orientation, reflect the vertical coordinate on each side of the same change of variables. The finite-cutoff rank domain is $y \le A_f(T)$ in the increasing orientation and $y \ge 1 - A_f(T)$ after reflection; zero-action axis-parallel segments complete the paths to the specified endpoints.

Finally let $T \to \infty$. On $0 \le x \le 1-\delta$, the profile has a positive lower bound, so a compactification $z = 1 - e^{-c_0 t}$ with $0 < c_0 < \frac{1}{2}\min_{x \le 1-\delta} f(x)$ gives a bounded continuous density. The omitted tail has vertical width $e^{-c_0 T}$, and Lemma 2.1 bounds its possible chain action by $O(e^{-c_0 T/2})$ in either orientation. Thus the finite-cutoff actions increase to $\sqrt{1-\delta}\,\mathcal{L}_\pm(f_\delta)$. On the rank side, the nonnegative truncated actions increase to the endpoint actions; taking the supremum over paths commutes with this increasing limit. This proves (90) without requiring a globally Lipschitz inverse at the singular endpoint.

Theorem 11.3 (Endpoint-degenerate Luce subsequence laws)**.** *Suppose there are $a_n > 0$ and a profile f satisfying (87) such that, for every $0 < \delta < 1$,*

$$\max_{1 \le i \le (1-\delta)n} \left| \frac{\theta_{n,i}}{a_n} - f(i/n) \right| \longrightarrow 0. \tag{91}$$

Assume also dyadic endpoint comparability (82). Then the monotone limits

$$\mathcal{L}_\pm^{\mathrm{sing}}(f) := \lim_{\delta \downarrow 0} \ell_f^\pm(\delta) = \lim_{\delta \downarrow 0} \mathcal{U}_{\rho_f^\pm}(1-\delta, 1) \tag{92}$$

exist and are finite. For $0 < \delta < 1$ they satisfy

$$0 \le \mathcal{L}_\pm^{\mathrm{sing}}(f) - \ell_f^\pm(\delta) \le \frac{a_K}{b_K}\sqrt{\delta}. \tag{93}$$

Moreover,

$$\frac{\mathrm{LIS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{L}_+^{\mathrm{sing}}(f), \tag{94}$$

$$\frac{\mathrm{LDS}(\sigma_n)}{\sqrt{n}} \longrightarrow \mathcal{L}_-^{\mathrm{sing}}(f) \tag{95}$$

in L^p for every $1 \le p < \infty$. For every $\varepsilon > 0$ there are $c_\varepsilon, C_\varepsilon > 0$ such that

$$\mathbb{P}\left(\max\left\{\left|\frac{\mathrm{LIS}(\sigma_n)}{\sqrt{n}} - \mathcal{L}_+^{\mathrm{sing}}(f)\right|, \left|\frac{\mathrm{LDS}(\sigma_n)}{\sqrt{n}} - \mathcal{L}_-^{\mathrm{sing}}(f)\right|\right\} > \varepsilon\right) \le C_\varepsilon e^{-c_\varepsilon\sqrt{n}} \tag{96}$$

for all sufficiently large n. Both convergences consequently hold almost surely under arbitrary couplings.

Proof. Write $L_n^+ = \mathrm{LIS}(\sigma_n)$ and $L_n^- = \mathrm{LDS}(\sigma_n)$. Let $N_{n,\delta} = \lfloor(1-\delta)n\rfloor$, and let $L_n^{(\delta),\pm}$ be the two chain lengths restricted to labels $1, \dots, N_{n,\delta}$. Reindexing these labels by $i/N_{n,\delta}$ changes the profile arguments by $O(n^{-1})$. Uniform continuity of f, condition (91), and Theorems 1.1 and 10.1 therefore give

$$\frac{L_n^{(\delta),\pm}}{\sqrt{n}} \longrightarrow \ell_f^\pm(\delta) \tag{97}$$

in L^p for every $1 \le p < \infty$, with fixed-deviation probability $O(e^{-c\sqrt{n}})$, for each fixed δ.

If $0 < \delta' < \delta$, then $L_n^{(\delta'),\pm} \ge L_n^{(\delta),\pm}$. Passing to the limits shows that $\ell_f^\pm(\delta)$ is nondecreasing as $\delta \downarrow 0$. Jensen's inequality and (83) with $m = n$ give

$$\mathbb{E}L_n^\pm \le \frac{a_K}{b_K}\sqrt{n}.$$

Hence both increasing limits in (92) are finite.

The number of labels outside the prefix is

$$m_{n,\delta} = n - N_{n,\delta} = \lceil \delta n \rceil.$$

Deleting these labels from a longest chain gives, for either sign,

$$0 \le L_n^\pm - L_n^{(\delta),\pm} \le R^\pm_{n,m_{n,\delta}}. \tag{98}$$

The same upper bound holds for $L_n^{(\delta'),\pm} - L_n^{(\delta),\pm}$ when $0 < \delta' < \delta$. Taking expectations, using (83) and Jensen's inequality, and then letting $n \to \infty$, gives

$$0 \le \ell_f^\pm(\delta') - \ell_f^\pm(\delta) \le \frac{a_K}{b_K}\sqrt{\delta}.$$

Letting $\delta' \downarrow 0$ proves (93).

For $1 \le p < \infty$, (85) and (98) imply

$$\left\| \frac{L_n^\pm - L_n^{(\delta),\pm}}{\sqrt{n}} \right\|_{L^p} \le C_{p,K}\sqrt{\delta + n^{-1}}.$$

The triangle inequality, (97), and (93) now prove (94)–(95), by taking $n \to \infty$ before $\delta \downarrow 0$.

For fixed $\varepsilon > 0$, choose δ so small that $(a_K/b_K)\sqrt{\delta} < \varepsilon/3$ and $a_K\sqrt{\delta} < b_K\varepsilon/6$. By (84),

$$\mathbb{P}\left(R^\pm_{n,m_{n,\delta}} > \frac{\varepsilon}{3}\sqrt{n}\right) \le \exp\left\{a_K\sqrt{\lceil\delta n\rceil} - \frac{b_K\varepsilon}{3}\sqrt{n}\right\} \le e^{a_K}e^{-b_K\varepsilon\sqrt{n}/6}.$$

Together with the prefix fixed-deviation estimate, the deterministic comparison, and (93), this proves (96), after a union bound over the two signs. The resulting probabilities are summable in n, so Borel–Cantelli gives almost sure convergence under every coupling. Finally, (90) identifies the variational limits in (92). □

11.1 Regularly varying endpoints

Corollary 11.4 (Sampled regularly varying profiles)**.** *Suppose f satisfies (87) and*

$$f(1-u) \sim u^{\gamma} L(1/u), \qquad u \downarrow 0, \tag{99}$$

where $\gamma > 0$ and L is slowly varying. If

$$\theta_{n,i} = a_n f\left(\frac{i}{n+1}\right)(1+\varepsilon_{n,i}), \qquad \max_i |\varepsilon_{n,i}| \longrightarrow 0, \tag{100}$$

then Theorem 11.3 applies to both increasing and decreasing subsequences.

Proof. Interior convergence follows from uniform continuity of f. In a dyadic terminal block, $u = (n+1-i)/(n+1)$ varies by a factor at most two. The uniform convergence theorem for regularly varying functions gives a uniform bound for the ratio of $f(1-u)$ at any two such points once u is small [4, Theorems 1.2.1 and 1.5.6]. A block not contained in this endpoint neighbourhood has all its u-values in a fixed compact subinterval of $(0,1]$, where continuity and positivity give the same bound. The relative errors in (100) change the ratio by a bounded factor for all sufficiently large n. Enlarging the constant to include the finitely many remaining arrays proves Definition 11.1. □

Corollary 11.5 (Both Sukhatme orientations)**.** *Let $f_{\downarrow}(x) = 1-x$, and set*

$$\mathcal{L}^{\downarrow}_{\mathrm{Suk}} = \mathcal{L}^{\mathrm{sing}}_{+}(f_{\downarrow}) = \lim_{\delta \downarrow 0} \mathcal{U}_{\rho_{\mathrm{Suk}}}(1-\delta, 1), \tag{101}$$

$$\mathcal{L}^{\uparrow}_{\mathrm{Suk}} = \mathcal{L}^{\mathrm{sing}}_{-}(f_{\downarrow}) = \lim_{\delta \downarrow 0} \mathcal{U}_{\rho^{-}_{\mathrm{Suk}}}(1-\delta, 1), \tag{102}$$

where ρ_{Suk} is given by (12) and $\rho^{-}_{\mathrm{Suk}}(x,y) = \rho_{\mathrm{Suk}}(x, 1-y)$. For Luce permutations $\sigma_n^{\downarrow}$ with rates $n-i+1$ and $\sigma_n^{\uparrow}$ with rates i,

$$\frac{1}{\sqrt{n}}\big(\mathrm{LIS}(\sigma_n^{\downarrow}), \mathrm{LDS}(\sigma_n^{\downarrow})\big) \longrightarrow \big(\mathcal{L}^{\downarrow}_{\mathrm{Suk}}, \mathcal{L}^{\uparrow}_{\mathrm{Suk}}\big),$$
$$\frac{1}{\sqrt{n}}\big(\mathrm{LIS}(\sigma_n^{\uparrow}), \mathrm{LDS}(\sigma_n^{\uparrow})\big) \longrightarrow \big(\mathcal{L}^{\uparrow}_{\mathrm{Suk}}, \mathcal{L}^{\downarrow}_{\mathrm{Suk}}\big).$$

These convergences hold in L^p for every $1 \le p < \infty$, with the fixed-deviation bound (96), and almost surely under arbitrary couplings. The constant $\mathcal{L}_{\mathrm{Suk}}$ in Corollary 1.3 is $\mathcal{L}^{\downarrow}_{\mathrm{Suk}}$.

Proof. Taking $f(x) = 1-x$ and $a_n = n+1$ in (100) gives exactly $\theta_{n,i} = n-i+1$; its dyadic endpoint constant is at most 2. Theorem 11.3 proves the first convergence and Corollary 1.3.

Let $r_n(i) = n+1-i$. Couple the increasing-rate clocks to the decreasing-rate clocks by $T^{\uparrow}_{n,i} = T^{\downarrow}_{n,r_n(i)}$. Then $\sigma_n^{\uparrow} = \sigma_n^{\downarrow} \circ r_n$. Reversal of the labels exchanges increasing and decreasing subsequences, which proves the second convergence with its stated constants and modes. □

Proposition 11.6 (Sukhatme truncation bounds)**.** *Let $f_{\downarrow}(x) = 1-x$ and $C_{\pm} = \mathcal{L}^{\mathrm{sing}}_{\pm}(f_{\downarrow})$, so that $C_{+} = \mathcal{L}^{\downarrow}_{\mathrm{Suk}}$ and $C_{-} = \mathcal{L}^{\uparrow}_{\mathrm{Suk}}$. For every $0 < \delta < 1$,*

$$\ell^{\pm}_{f_{\downarrow}}(\delta) \le C_{\pm} \le \frac{\ell^{\pm}_{f_{\downarrow}}(\delta)}{1-\sqrt{\delta}}. \tag{103}$$

Proof. Consider the rates $n, n-1, \dots, 1$, and put $N = \lfloor (1-\delta)n \rfloor$ and $m = n - N = \lceil \delta n \rceil$. The final m clocks have exactly the rates $m, m-1, \dots, 1$. For either orientation, write $L_n^{\pm}$ for the full chain length, $P_{n,\delta}^{\pm}$ for the prefix chain length, and $S_{n,m}^{\pm}$ for the suffix chain length. Deterministically,

$$P_{n,\delta}^{\pm} \le L_n^{\pm} \le P_{n,\delta}^{\pm} + S_{n,m}^{\pm},$$

and $S_{n,m}^{\pm}$ has the same distribution as $L_m^{\pm}$. Take expectations and divide by $\sqrt{n}$. The prefix limit (97) and the L^1 convergence in Theorem 11.3 give

$$\ell_{f_\downarrow}^{\pm}(\delta) \le C_{\pm} \le \ell_{f_\downarrow}^{\pm}(\delta) + \sqrt{\delta}\, C_{\pm}.$$

Rearrangement proves (103). □

Proposition 11.7 (An explicit Sukhatme LIS lower bound)**.** *The constant for decreasing Sukhatme rates satisfies*

$$\mathcal{L}_{\mathrm{Suk}}^{\downarrow} > 3.0602. \tag{104}$$

More generally, for every $a > 0$,

$$\mathcal{L}_{\mathrm{Suk}}^{\downarrow} \ge J(a) := 4\sqrt{a}\, e^{-a/2} \int_0^1 e^{au^2/2}\, du. \tag{105}$$

Proof. For the raw-clock density $(1-x)e^{-(1-x)t}$, consider the increasing curve

$$t_a(x) = \frac{ax}{1-x}, \qquad 0 \le x < 1.$$

Restrict it to $x \le 1-\delta$, transform to rank coordinates, and append a zero-action vertical segment to rank 1. This gives an admissible path for $\ell_{f_\downarrow}^{+}(\delta)$. Since $t_a'(x) = a/(1-x)^2$, its actions increase as $\delta \downarrow 0$ to

$$2\sqrt{a} \int_0^1 \frac{e^{-ax/2}}{\sqrt{1-x}}\, dx = 4\sqrt{a}\, e^{-a/2} \int_0^1 e^{au^2/2}\, du,$$

where the equality uses $u = \sqrt{1-x}$. This proves (105).

Take $a = 7/4$ and put

$$A_{13} = \sum_{j=0}^{13} \frac{(-7/8)^j}{j!}, \qquad B_{12} = \sum_{k=0}^{12} \frac{(7/8)^k}{k!(2k+1)}.$$

Taylor's formula and positivity of the omitted integral terms give

$$0 < A_{13} < e^{-7/8}, \qquad B_{12} < \int_0^1 e^{7u^2/8}\, du.$$

Squaring verifies $\sqrt{7} > 26457513/10^7$, and direct rational arithmetic gives

$$J(7/4) > \frac{2 \cdot 26457513}{10^7} A_{13} B_{12} > \frac{15301}{5000} = 3.0602.$$

Together with (105), this proves (104). □

The singularity has a universal local form on comparable endpoint scales. The next statement is independent of the probabilistic truncation argument.

Proposition 11.8 (Regularly varying corner kernel)**.** *In addition to (99), suppose $r(u) = f(1-u)$ is eventually C^1, strictly increasing, and $ur'(u)/r(u) \to \gamma$. Put*

$$G_\gamma = \Gamma(1 + 1/\gamma), \qquad \psi_\gamma(s) = \gamma G_\gamma^\gamma s^\gamma e^{-G_\gamma^\gamma s^\gamma}.$$

Then, locally uniformly for $s \in (0, \infty)$,

$$v\,\rho_f(1 - sv, 1 - v) \longrightarrow \psi_\gamma(s) \qquad (v \downarrow 0). \tag{106}$$

Proof. Let $S(t) = 1 - A_f(t) = \int_0^1 e^{-tf(x)}\,dx$ and let $h(t)$ be the asymptotic inverse scale determined by $r(h(t)) \sim 1/t$. Karamata's Laplace theorem and the monotone density theorem give

$$S(t) \sim G_\gamma h(t), \qquad B_f(t) = -S'(t) \sim \frac{S(t)}{\gamma t};$$

see [4, Sections 1.5 and 4.12]. If $v = S(t)$ and $u = sv$, regular variation yields

$$tr(u) \longrightarrow G_\gamma^\gamma s^\gamma,$$

locally uniformly for $s \in (0, \infty)$. Since $\rho_f(1 - u, A_f(t)) = r(u)e^{-tr(u)}/B_f(t)$, substitution gives (106). □

For the Sukhatme profile $\gamma = 1$, this reads

$$v\,\rho_{\mathrm{Suk}}(1 - sv, 1 - v) \longrightarrow se^{-s}.$$

12 Further questions

The first-order and localization laws leave several natural problems.

Conjecture 12.1 (KPZ fluctuations with exact centering)**.** *Assume an exactly sampled C^3 positive profile and a unique smooth nondegenerate maximizing path separated from the boundary. Then there is $s_f > 0$ such that, writing χ_2 for a random variable with the GUE Tracy–Widom law,*

$$\frac{\mathrm{LIS}(\sigma_n) - \mathbb{E}\,\mathrm{LIS}(\sigma_n)}{s_f n^{1/6}} \Rightarrow \chi_2 - \mathbb{E}\chi_2. \tag{107}$$

Equivalently, one expects

$$\frac{\mathrm{LIS}(\sigma_n) - \mathcal{L}_+(f)\sqrt{n}}{s_f n^{1/6}} \Rightarrow \chi_2$$

whenever the exact finite-n model has no additional deterministic $n^{1/6}$-order centering term. In that case

$$\mathbb{E}\,\mathrm{LIS}(\sigma_n) = \mathcal{L}_+(f)\sqrt{n} + s_f(\mathbb{E}\chi_2)n^{1/6} + o(n^{1/6}).$$

The exact-centering qualification is essential. The first-order hypothesis $\max_i |\varepsilon_{n,i}| \to 0$ allows deterministic profile errors large enough to shift the LIS by more than $n^{1/6}$. Centering by the expectation avoids that ambiguity, but necessarily centers the limiting Tracy–Widom variable as well. The uniform case is the Baik–Deift–Johansson theorem [2]. Different limits may occur at caustics, for boundary-touching maximizers, or when the maximizing path is not unique.

Conjecture 12.2 (Row-stratified Greene–RSK surface)**.** *For a continuous row density q, the complete rescaled Young diagram of the row-stratified word has a deterministic variational limit. Its Greene invariants are described by the corresponding optimization over disjoint families of monotone paths. Under the rank-coordinate map, this yields a full RSK limit surface for independent-score and regular Luce rankings.*

This would extend the locally uniform permutation theory of Sjöstrand [17] and the permuton-based work of Dubach [10, 11] to deterministic one-point-per-fibre arrays.

Conjecture 12.3 (Rapidly varying endpoint transition)**.** *For geometric Sukhatme weights*

$$\theta_{n,i} = \alpha^{n-i+1}, \qquad \alpha > 1,$$

there is a constant $v(\alpha) \in (0,1)$ such that

$$\frac{\mathrm{LIS}(\sigma_n)}{n} \longrightarrow v(\alpha).$$

More generally, the LIS exponent for endpoint arrays outside the dyadically comparable class is determined by the number of labels occupying each multiplicative rate scale.

The geometric model is the simplest regime in which each successive block contains only $O(1)$ labels per rate decade. It should therefore lie in a linear rather than a Hammersley $\sqrt{n}$ class. Intermediate arrays, such as rates varying like $\exp\{\beta_n i/n\}$ with $\beta_n \to \infty$, may exhibit a crossover governed by the effective number of labels per logarithmic rate scale.

Conjecture 12.4 (Rough row profiles)**.** *Theorem 2.2 remains valid for bounded measurable row densities q under a suitable Riemann-approximability hypothesis, with the upper semicontinuous relaxation of the Hammersley action when q has genuine discontinuities.*

Continuity is used in the proof only to make the within-block oscillation vanish uniformly and to identify the block functional. This suggests extensions to regulated and piecewise continuous profiles and, with a refined exceptional-block argument, to L^1 approximation.

Conjecture 12.5 (Moderate deviations and transversal fluctuations)**.** *In the unique interior-geodesic regime, the fixed-deviation estimate of this paper admits a uniform moderate-deviation interpolation down to the $n^{1/6}$ longitudinal and $n^{-1/6}$ transversal scales, with the usual KPZ exponents and profile-dependent curvature constants.*

Remark 12.6 (Scope of the proved concentration). The bounds proved here concern fixed deviations on the $\sqrt{n}$ first-order scale. They do not by themselves imply Tracy–Widom fluctuations, moderate deviations, or KPZ transversal exponents.

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